

Barcelona School of Economics (BSE)

Advanced Techniques in Applied Economics (Professor Piotr Zwiernik)

Estimating latent factors in dynamic models

The Kalman filter under misspecification

Table of contents

Abstract	3
1 The Stock and Watson model	4
1.1 Latent factor estimation	5
1.2 Mapping the latent factor to GDP growth	6
2 Breaking Gaussianity	8
2.1 Gaussian distribution	9
2.2 Student-t distribution	10
2.3 Skew-t distribution	11
2.3.1 Skew-t with increasing skewness	12
2.4 Monte Carlo simulation and efficiency loss under non-Gaussian data	13
3 Two-factor model	15
3.1 Parallel analysis to determine the number of factors	15
3.2 Introducing price-side indicators	16
3.3 Re-estimating with the correct specification	19
4 Conclusion	21
Bibliography	22

Abstract

In a 1988 working paper, Stock and Watson proposed a framework to estimate and forecast short-run U.S. economic activity. Their model built on the premise that co-movements across multiple macroeconomic variables can be captured by a single, unobserved latent variable. Specifically, they hypothesized that their four chosen indicators form a dynamic single-factor model in which the latent factor represents the state of the business cycle and can therefore be mapped to Gross Domestic Product (GDP) growth Stock and Watson (1988).

This project empirically tests the Kalman filter under two different forms of misspecification, departing from the original Stock and Watson assumptions. We begin in Section 1 by replicating the model on more recent data to establish a performance benchmark, confirming that the original model yields a latent factor that continues to track GDP growth well, with an R^2 of 0.763 and MSE of 0.276.

Next, in Section 2 we assess the robustness of the Kalman filter to violation of the Gaussianity assumption. Since macroeconomic data frequently exhibits heavy tailed and asymmetric behaviors, we simulate economies under progressively more elliptical and skewed distributions. We find that convergence to the true latent factor is faster under Gaussian errors and slows as distributions become more elliptical or skewed. However, the Kalman filter continues to recover the true factor in large samples across all specifications.

Finally, in Section 3 we expand the number of indicators, including price-side variables and therefore introducing a second latent factor into the data-generating process. Estimating a single-factor model on this augmented dataset, we find that the Kalman filter recovers only the real-activity factor, assigning loadings close to zero to all price indicators. After re-estimating the model under the correct two-factor specification, the Kalman filter successfully recovers both the real-activity factor and price-pressure factor, yielding an R^2 of 0.761 against GDP growth and 0.732 against PCEPI inflation.

1 The Stock and Watson model

We first replicate the model proposed by Stock and Watson. Their model assumes that the co-movements of four macroeconomic indicators are driven by a single, unobserved latent factor which represents the overall business cycle. The included indicators are:

- IPMAN (Industrial Production: Manufacturing).
- CMRMTSPL (Real Manufacturing and Trade Industries Sales).
- PAYEMS (Total Nonfarm Payrolls).
- W875RX1 (Real Personal Income Excluding Current Transfer Receipts).

These series are FRED equivalents of the original Stock and Watson indicators, following the implementation of Fulton (2015) as part of the *statsmodels* library (Seabold and Perktold 2010). Indeed, each series captures the same underlying dimension of real economic activity as the original indicators included in the Stock and Watson paper: industrial output (*INDPRO*), goods and services demand (*CMRMTSPL*), labor market conditions (*PAYEMS*), and household earned income (*W875RX1*). Therefore, results should be interpreted as a conceptual replication rather than a numerical one, and differences in estimated loadings are mostly attributable to the series substitutions.

To capture the cyclic pattern of business cycles, both the factor and the error follow an $AR(2)$ process. Visually, the model can be represented by the graphical model represented in Figure 1.

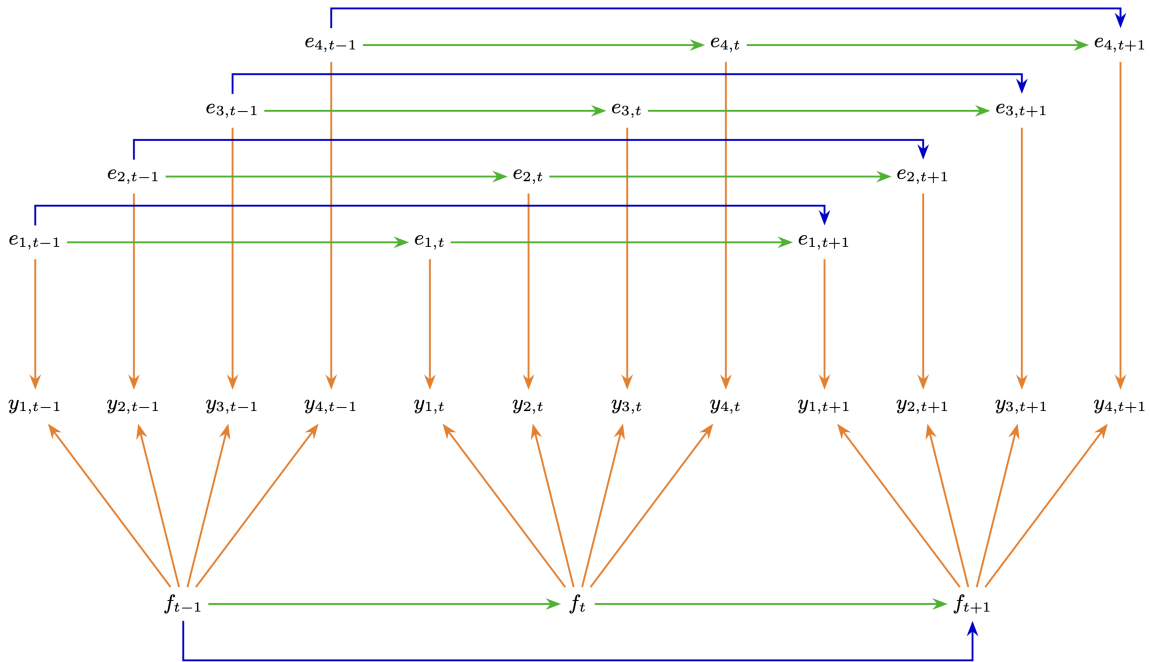


Figure 1: Stock & Watson dynamic factor model (Stock and Watson 1988).

Formally, the model is represented by the following structural equations:

- The measurement equation: $y_t = \lambda f_t + e_t$
- The factor transition equation: $f_t = \phi_1 f_{t-1} + \phi_2 f_{t-2} + w_t$, $w_t \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, 1)$
- The error transition equation: $e_t = \psi_1 e_{t-1} + \psi_2 e_{t-2} + \epsilon_t$, $\epsilon_t \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma_i^2)$

Where:

- y_t is the (4×1) vector of normalized log-differences of the indicators at time t
- λ is the (4×1) vector of coefficients
- f_t is the (1×1) factor scalar at time t
- e_t is the (4×1) vector of errors at time t

Notice that this model assumes two conditional independence statements. First, the four indicators are cross-sectionally conditionally independent given the latent factor, meaning that once the common factor is known, knowing one indicator provides no additional information about the others. Formally:

$$y_{i,t} \perp\!\!\!\perp y_{j,t} \mid f_t \quad \forall i \neq j$$

Second, since both the factor and the errors follow an AR(2) process, given the state vector h_t current indicators are conditionally independent of all future and past observations. Formally:

$$y_t \perp\!\!\!\perp y_s \mid h_t \quad \forall t \neq s$$

Where:

$$h_t = (f_t, f_{t-1}, e_{1,t}, e_{1,t-1}, \dots, e_{n,t}, e_{n,t-1})$$

1.1 Latent factor estimation

Given that the model is linear and that shocks w_t and ϵ_t are assumed to be Gaussian and independent, the optimal estimator for the latent factor is the Kalman filter (Kalman 1960). At each period t , the filter proceeds in two steps. In the prediction step, it forecasts the state vector h_t from the previous estimate using the transition equation. In the update step, after observing y_t , it computes the prediction error and corrects the state estimate:

$$\hat{h}_{t|t} = \hat{h}_{t|t-1} + K_t e_{t|t-1}, \quad K_t = P_{t|t-1} H^\top (H P_{t|t-1} H^\top + R)^{-1}$$

The Kalman gain K_t shrinks when observations are noisy, causing the filter to rely more heavily on its own prediction, while it grows when observations are reliable, weighting more heavily the new data and therefore the prediction error. For a full derivation, see (Hamilton 1994, Ch. 13).

The Kalman filter, however, requires the errors of the measurement equation e_t to be independent white noise, unlike the AR(2) specification of the Stock and Watson model. To overcome this limitation, we take the AR(2) errors out of the measurement equation and include them into the latent state vector h_t , alongside the current and lagged values of the factor. With this structure, the F matrix handles the AR(2) dynamics of both the factor and the error and the model reduces to the following state-space system:

- The measurement equation: $y_t = H h_t$

- The factor transition equation: $h_t = Fh_{t-1} + v_t$, $v_t \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma)$

The vector of optimal parameters is estimated via Gaussian Maximum Likelihood, where:

$$\theta = [\lambda_1, \dots, \lambda_4, \phi_1, \phi_2, \psi_{11}, \dots, \psi_{42}, \sigma_1^2, \dots, \sigma_4^2]'_{(18 \times 1)}$$

1.2 Mapping the latent factor to GDP growth

After estimating the latent factor using the Kalman filter, we rescale it to make it interpretable in terms of actual economic activity. To do so, we regress GDP onto the estimated factor by Ordinary Least Squares (OLS). This allows us to plot the estimated latent factor against GDP growth and calculate in-sample performance metrics, such as the coefficient of determination (R^2) and the Mean Squared Error (MSE).

As shown in Figure 2, empirical results demonstrate the strong explanatory power of the model proposed by Stock and Watson. The mapped factor explains approximately 76% of the total variance of actual quarterly GDP growth ($R^2 = 0.763$). Moreover, the model yields an in-sample MSE of 0.276, translating to a Root Mean Squared Error (RMSE) of roughly 0.526 percentage points. This RMSE might seem elevated, but it is important to note that our dataset includes periods of extreme macroeconomic volatility, such as the 2008 and COVID-19 crises, which inflate the error metrics.

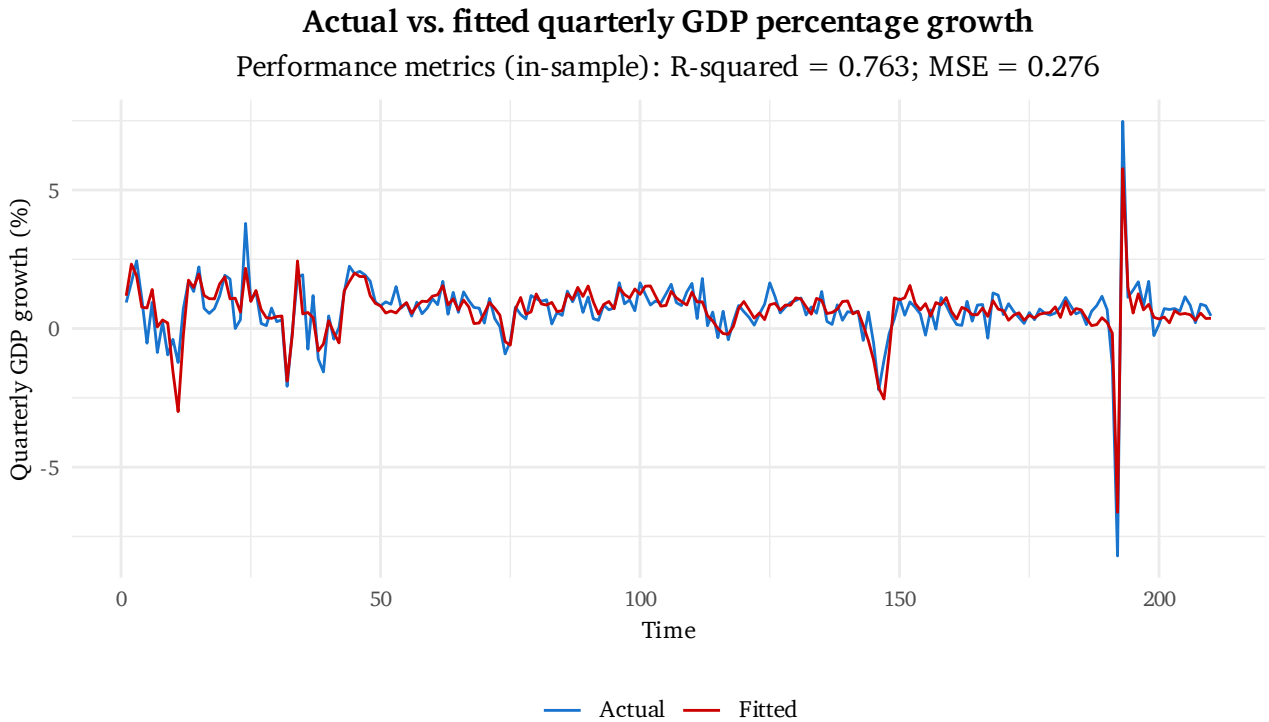


Figure 2: Fitted vs. actual GDP quarterly growth, original Stock and Watson model.

As reported in Table 1, all four indicators load positively on the estimated factor, in decreasing order of magnitude: *IPMAN* ($\hat{\lambda} = 0.926$), *PAYEMS* ($\hat{\lambda} = 0.782$), *CMRMTSPL* ($\hat{\lambda} = 0.711$), and *W875RX1* ($\hat{\lambda} = 0.525$). The results are consistent with the original S&W estimates, where all indicators had

positive loading and industrial production also carried the largest loading (0.717) (Stock and Watson 1988).

Table 1: Maximum likelihood estimates of the Stock and Watson model

Parameter	Coincident Indicator			
	IPMAN	CMRMTSPL	PAYEMS	W875RX1
λ_i	0.926 (0.031)	0.711 (0.031)	0.782 (0.035)	0.525 (0.036)
ψ_{1i}	0.076 (0.134)	-0.324 (0.045)	-0.020 (0.048)	-0.143 (0.041)
ψ_{2i}	0.086 (0.133)	-0.135 (0.043)	0.103 (0.050)	-0.073 (0.041)
σ_i^2	0.081 (-0.023)	0.393 (0.026)	0.371 (0.027)	0.700 (0.042)

Standard errors in parentheses.

2 Breaking Gaussianity

The dynamic factor model proposed by Stock and Watson explicitly assumes that all structural shocks are purely Gaussian, which grants it the property of being the minimum mean square error (MMSE) estimator of the latent factor, as proven by Hamilton (1994, sec. 13.2).¹ However, economic cycles frequently feature sudden and massive shocks (e.g. the COVID-19 crisis), hence macroeconomic indicators are rarely normally distributed. Instead, they tend to exhibit heavy-tailed behavior. As illustrated in the Quantile-Quantile (Q-Q) plot below (Figure 3), the distribution of our indicators indeed deviates from normality, confirming the presence of fat tails.

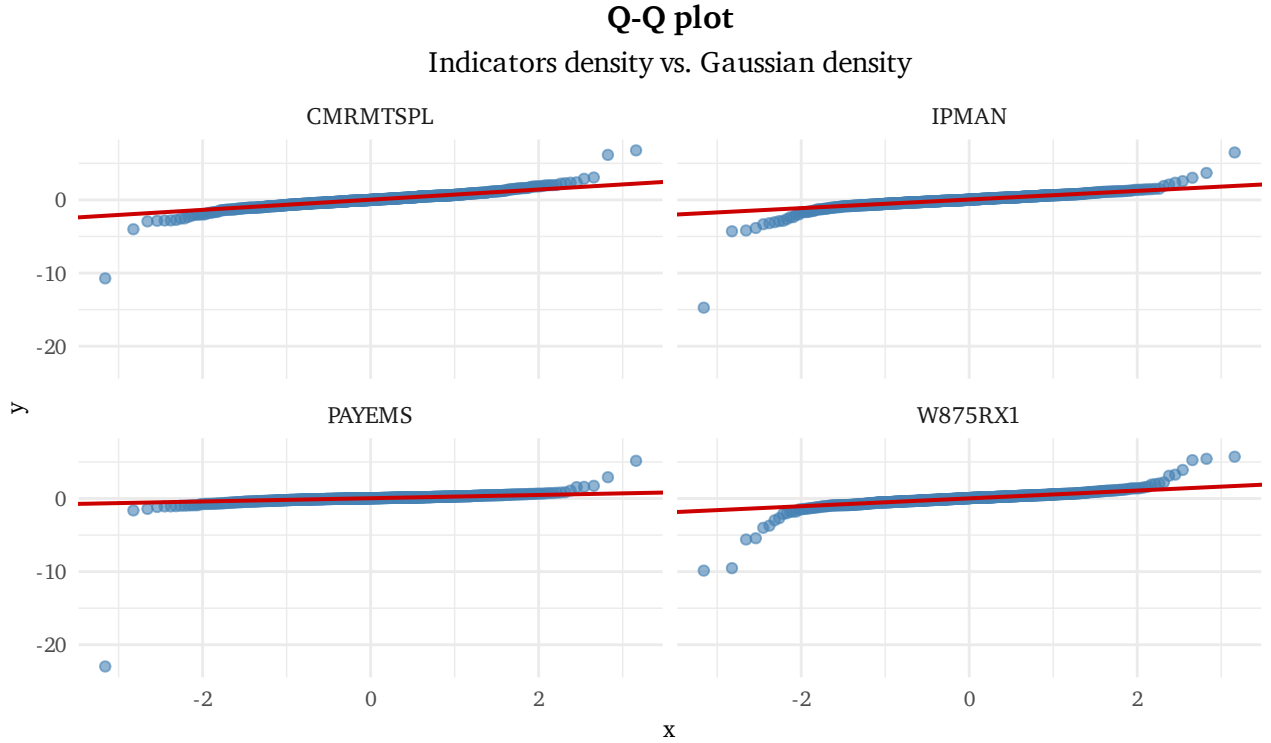


Figure 3: Normality diagnostics for macro indicators.

Despite the violation of the normality assumption, the Kalman filter extracts a consistent estimate of the latent factor. Indeed, when the parameters are estimated via Gaussian quasi-maximum likelihood, consistency of the parameter estimates is preserved under non-Gaussianity, though with loss of efficiency in finite samples as the distribution moves away from Gaussianity and gets increasingly elliptical (White 1982).

The reason for this is that under a misspecified likelihood, the Kalman gain is no longer optimal; therefore, the filter requires more observations to achieve the same precision as it would under correctly specified Gaussian shocks. This efficiency loss is particularly relevant for macroeconomic data, which, with fewer than 700 observations, can never be confidently characterized as truly Gaussian, and empirically exhibits heavy-tailed behavior consistent with a student-t distribution as shown in *Table 1* of

¹This follows from the fact that the model is linear and that shocks w_t and ϵ_t are assumed to be Gaussian and independent; the conditional expectation $\mathbb{E}[h_t|y_{1:t}]$ minimizes mean squared error among all measurable functions of $y_{1:t}$. See Hamilton (1994, sec. 4.1) for a standard proof.

(J. Geweke 1993).² We evaluate this by simulating the S&W data generating process under three distributional assumptions of increasing complexity: Gaussian, student-t, and skew-t, the latter of which introduces nonzero skewness via a slant parameter $\alpha \neq 0$.

To better understand this phenomenon, we generate synthetic data based on our coefficient estimates derived from Section 1.2 that increasingly diverge from normality. We first understand the behavior of the efficient parameters derived under Gaussian distribution in Section 2.1, after which we move into a typical finite-sample distribution of student-t in Section 2.2, and finally we explore more complex distributions such as skew-t with varying degrees of skew in Section 2.3 to understand where exactly the filter breaks down.

2.1 Gaussian distribution

To simulate Gaussian data into our Kalman filter, we generate a structured array of various sample sizes to evaluate how the estimator performs in finite samples and what happens to its performance as $T \rightarrow \infty$.

For each sample size, we generate a true latent factor following an AR(2) process with coefficient values of $\hat{\phi}_1 = 0.240$ and $\hat{\phi}_2 = -0.073$, estimated from the S&W model in Section 1.1, with Gaussian shocks by assumption. We then construct the observable data by multiplying the true factor by the estimated factor loadings from Section 1.1 and adding normally distributed idiosyncratic errors, replicating the structure of the S&W measurement equation. The simulated observables are normalized and fed into the Kalman filter, and the correlation between the estimated factor $\hat{f}_t$ and the true factor f_t is recorded across all sample sizes.

We first run a single iteration across sample sizes generated from `seq(10, 10000, by = 50)`, such that the lower bound captures realistic finite sample sizes representative of macroeconomic datasets, while the upper bound approaches the asymptotic nature of a dataset, where the property of consistency is expected to hold.

We see from Figure 4 that factor recovery under Gaussian shocks is stable across all sample sizes, with correlation between the estimated and true factor converging to approximately 0.7787 as T grows. The relatively high recovery even at small sample sizes is consistent with the Kalman filter being the minimum variance estimator under correctly specified Gaussian shocks, confirming that no efficiency loss is present when the distributional assumption holds.

²J. Geweke (1993) specifically tests on fourteen macroeconomic indicators including Industrial Production, Employment, and Real GNP, finding overwhelmingly positive evidence against normality at decreasing degrees of freedom ($\nu = 20, 10, 5, 3, 1$), in favor of student-t errors at low-to-moderate degrees of freedom. This motivates our decision to use student-t and naturally skew-t as distributional assumptions away from normality, with our degrees of freedom chosen as $\nu = 5$ in the simulations.

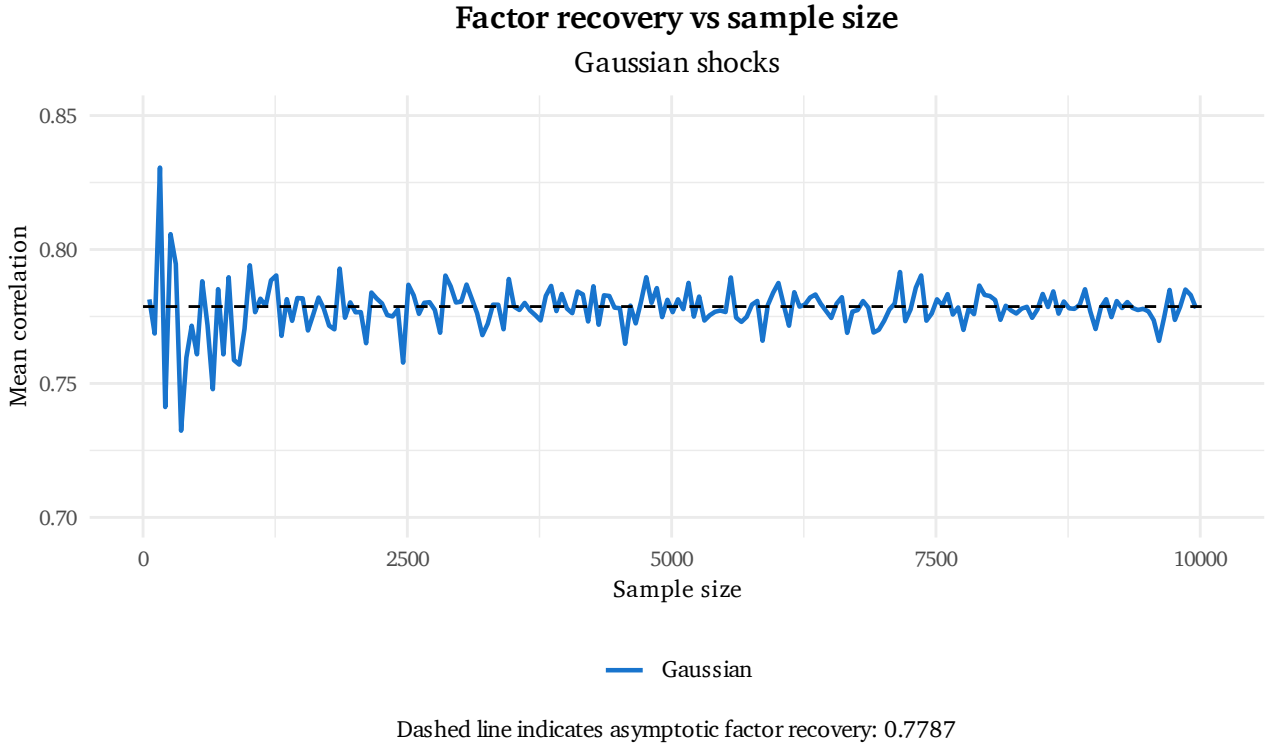


Figure 4: Factor recovery vs sample size under Gaussian shocks.

2.2 Student-t distribution

To break the assumption of Gaussianity, we simulate the S&W data generating process under a student-t distribution, which is well documented as a realistic characterization of finite sample macroeconomic data (J. Geweke 1993).

Theoretically, given the QMLE consistency property, we should expect the Kalman filter to remain consistent in its recovery of the latent factor, meaning the estimates converge to the true factor as $T \rightarrow \infty$. However, we expect to observe an efficiency loss in finite samples, as the Kalman gain is derived under the Gaussian likelihood and is therefore suboptimal when the true distribution has heavier tails (White 1982; Durbin and Koopman 2012). Here we select the degrees of freedom to be $\nu = 5$, as supported by empirical results from *Table 1* in (J. Geweke 1993), as the posterior odds ratio strongly support student-t distribution at this level. Lower values of ν would generate extreme draws too frequently to match the empirical behavior of macroeconomic data, where shocks like the COVID-19 disruption are severe but rare. The choice $\nu = 5$ is also the lowest degree of freedom at which both the variance and kurtosis of the student-t distribution remain finite, satisfying the regularity conditions required for QMLE consistency (i.e. finite fourth moments and non-zero variance; see *Theorem 3.3* in (White 1982)).

As illustrated in Figure 5, the Kalman filter remains consistent under student-t shocks, with factor recovery converging to approximately 0.7787 as $T \rightarrow \infty$, in line with the Gaussian baseline from Section 2.1. However, the trajectory is notably more volatile across sample sizes, particularly in the finite sample realm where the correlation fluctuates significantly before stabilizing. This is consistent with the theoretical prediction that the Kalman filter loses efficiency under heavy-tailed distributions, as the

misspecified Gaussian likelihood produces a suboptimal Kalman gain that requires more observations to average out the influence of extreme draws.

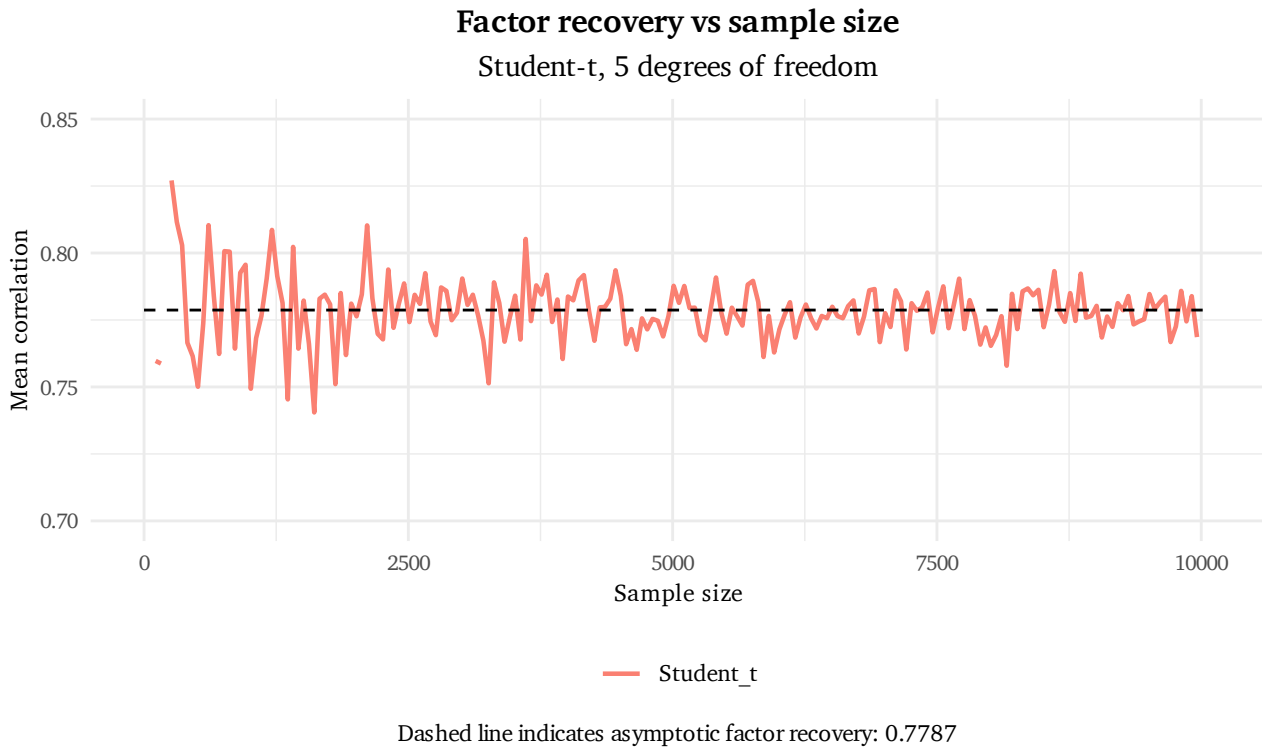


Figure 5: Factor recovery vs sample size under Student-t shocks.

In other words, the fat tails of the student-t distribution introduce large idiosyncratic shocks that the filter struggles to separate from the common factor signal at small T , resulting in noisier factor recovery relative to the Gaussian case. Despite this finite-sample volatility, the asymptotic recovery converges to the same value as the Gaussian baseline, confirming that the QMLE consistency property holds even under substantial deviations from normality.

2.3 Skew-t distribution

For a more complex distributional assumption, we simulate the S&W data generating process under a skew-t distribution, which extends the student-t by introducing asymmetry via a nonzero slant parameter $\alpha \neq 0$.

Specifically, we set $\alpha = 5$ which induces a positive skew, meaning the distribution has a longer right tail relative to the symmetric student-t case. Note that our simulated factor shocks are generated as follows:

```
factor_shocks <- rst(tt, xi=0, omega=1, alpha=5, nu=5)
factor_shocks <- factor_shocks - mean(factor_shocks)
```

The factor shocks are recentered by subtracting the sample mean, as the skew-t distribution has a nonzero mean when $\alpha \neq 0$, which would otherwise violate the zero mean assumption of the Kalman filter and introduce a systematic bias into the factor estimates.

Indeed, from our results in Figure 6 we see that the Kalman filter remains consistent under skew-t shocks, with factor recovery converging to approximately 0.7787 as $T \rightarrow \infty$, in line with both the Gaussian and student-t baselines. This confirms the QMLE consistency result, as the estimator recovers the true latent factor regardless of the distributional assumption placed on the shocks.

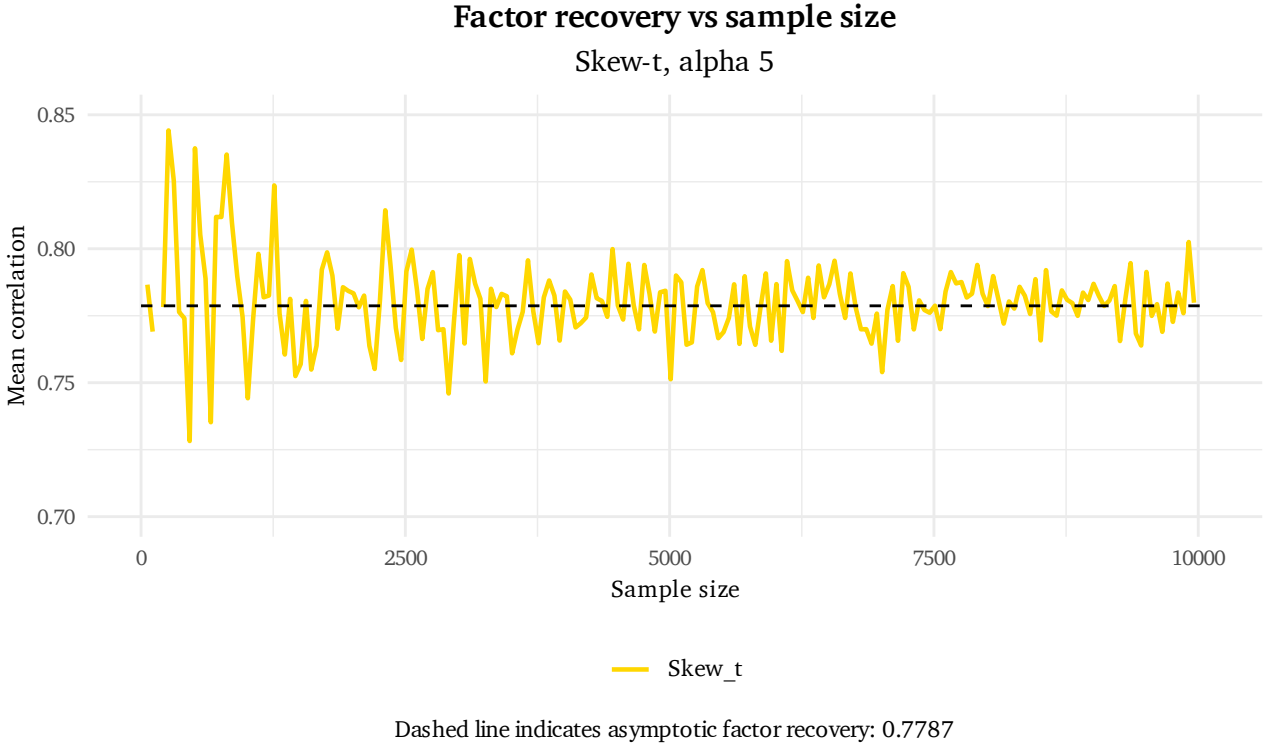


Figure 6: Factor recovery vs sample size under Skew-t shocks.

However, the trajectory is notably more volatile relative to both the Gaussian and student-t cases, with correlation fluctuating between 0.75 and 0.80 even at large sample sizes. This persistent volatility reflects the additional efficiency cost introduced by the nonzero skewness of the shock distribution, as the positive slant parameter $\alpha = 5$ generates asymmetric extreme draws that the Kalman filter, operating under a misspecified Gaussian likelihood, struggles to average out even at large T .

2.3.1 Skew-t with increasing skewness

The skew-t result with $\alpha = 5$ recovers the factor surprisingly well at a large sample size. An interesting extension would be to understand at exactly what skew parameter value does the skew-t become too extreme to consistently recover the factor. To explore this, we again run our simulation on skew-t generated data, this time with increasing, positive skew values: $\alpha = 1, 3, 5, 10, 20, 50$. We conduct a single iteration of the MC simulation for our array of values spanning from $n = 10, \dots, 10000$.

From our results in Figure 7, we see that even at extreme skews of $\alpha = 50$ the asymptotic factor recovery is essentially invariant to the choice of skewness parameter. The volatility of the convergence in finite samples is similarly comparable across alpha values, with no clear monotonic relationship between skewness and convergence speed. This confirms the QMLE consistency property in its strongest form, wherein the Kalman filter recovers the latent factor regardless of the magnitude of the skewness in

the shock distribution (provided the regularity conditions on finite moments hold, which they do for $\nu = 5$).

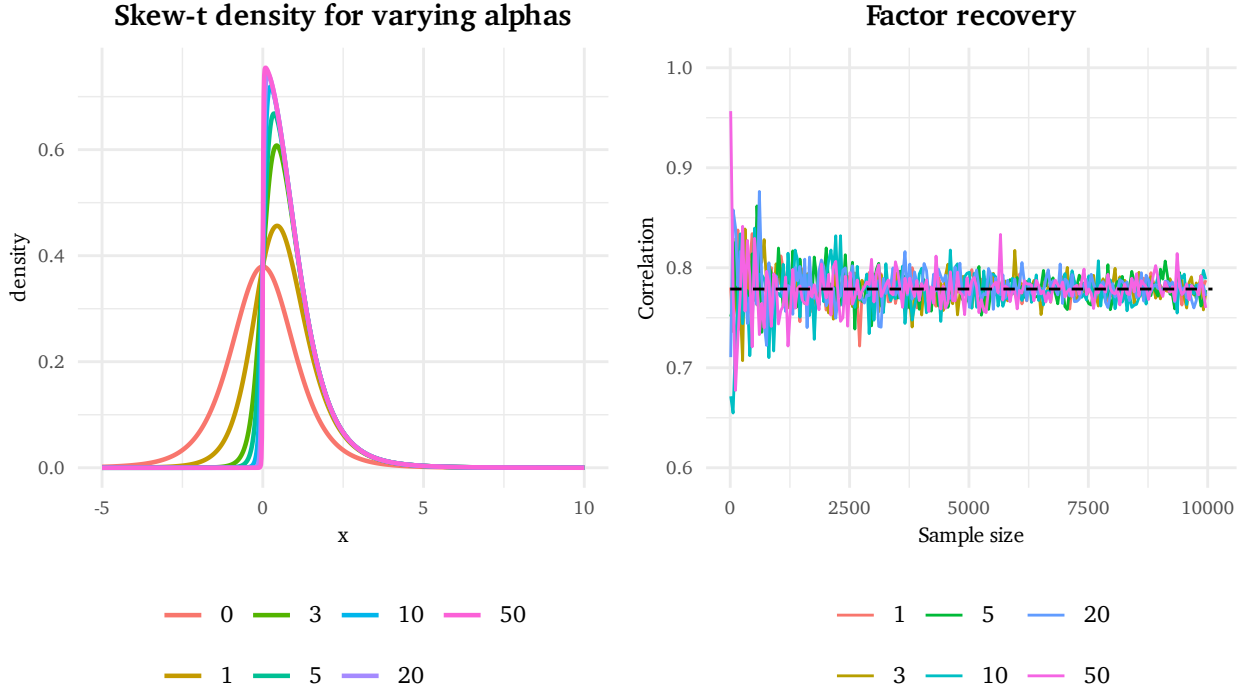


Figure 7: Skew-t distributions across varying alphas.

This implies that the direction of distributional misspecification matters less than its presence. Whether the shocks are symmetric heavy-tailed (student-t, Section 2.2) or asymmetric heavy-tailed (skew-t at any level of α), the asymptotic recovery is preserved for the Stock and Watson dynamic factor model. The cost of misspecification is paid in finite sample volatility, not in the limit.

2.4 Monte Carlo simulation and efficiency loss under non-Gaussian data

As illustrated in Figure 8, the three distributions converge to the same asymptotic factor recovery correlation of approximately 0.7787 as $T \rightarrow \infty$, directly confirming the QMLE consistency result. At small sample sizes, all three distributions exhibit greater volatility and slightly lower recovery, with the skew-t and student-t lines deviating more from the Gaussian baseline, consistent with the theoretical prediction of finite sample efficiency loss under non-Gaussianity.

Notably, both the student-t and skew-t distributions exhibit greater volatility relative to the Gaussian baseline, particularly at small sample sizes, with the skew-t remaining the most volatile throughout. It should be noted that the y-axis range is intentionally narrow to highlight these differences, and as reported in Table 2, the absolute magnitude of the fluctuations is small in practice, with standard deviations of 0.0254 and 0.0304 for the student-t and skew-t, respectively. Nevertheless, all three distributions converge to the same asymptotic recovery value of 0.7787, confirming the QMLE consistency result regardless of the distributional assumption.

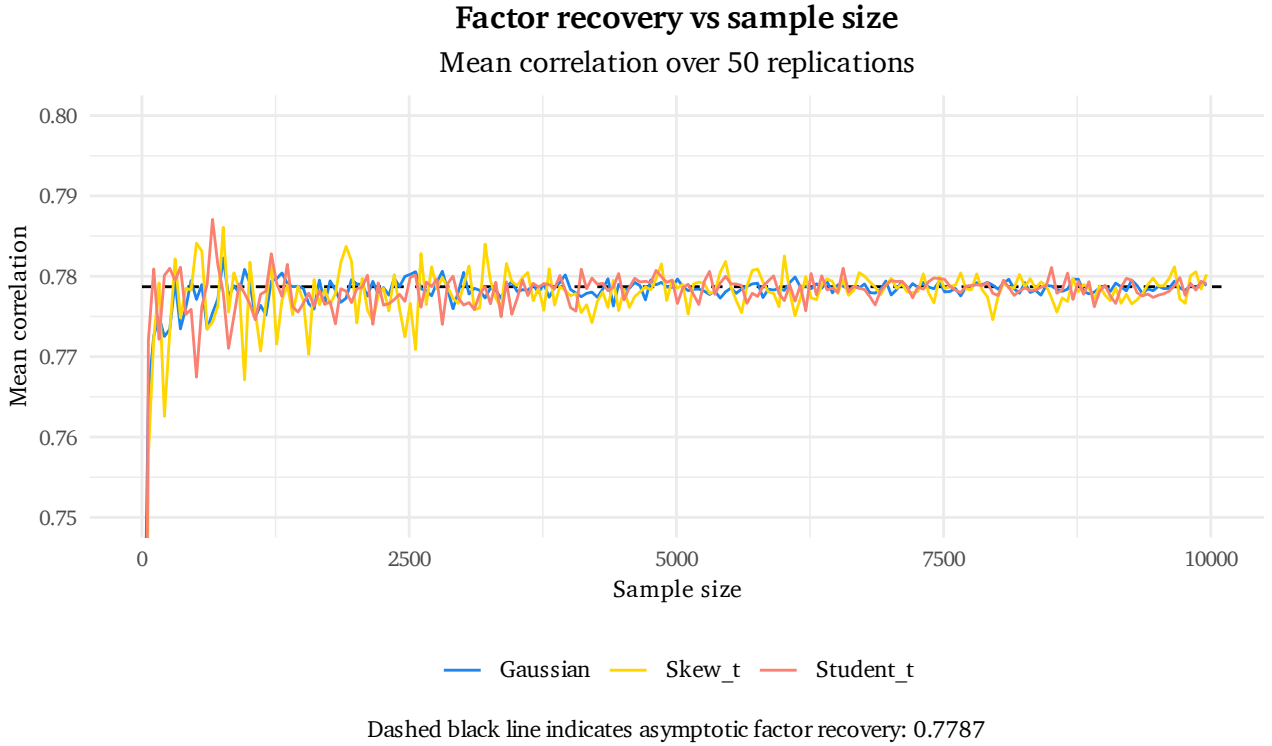


Figure 8: Factor recovery vs sample size, averaged across 50 MC simulations.

To show the efficiency loss under non-Gaussian distributions, we run 200 replications of the Monte Carlo simulation at the empirical sample size of $T = 633$, corresponding to the number of observations in our FRED dataset, and compute the variance of factor recovery across replications.

As reported in Table 2, the Gaussian estimator achieves the lowest variance of 0.000243, confirming its optimality under correctly specified shocks. The student-t estimator exhibits a variance of 0.000644, roughly three times that of the Gaussian case, while the skew-t estimator yields the highest variance of 0.000924, approximately four times the Gaussian baseline. Importantly, the mean recovery is nearly identical across all three distributions at approximately 0.7787, confirming that the estimator remains consistent regardless of the distributional assumption.

Table 2: Factor recovery: mean and variance across 200 replications.

Distribution	Mean	Variance	SD
Gaussian	0.776963	0.000243	0.015573
Student-t	0.778177	0.000644	0.025372
Skew-t	0.777062	0.000924	0.030404

Together, these results provide formal evidence of the consistency without efficiency property of the Gaussian quasi-maximum likelihood Kalman filter: the estimator converges to the correct value under all three distributions, but requires significantly more observations to do so with the same precision when shocks are non-Gaussian.

3 Two-factor model

In this chapter we empirically examine how the Kalman filter behaves when the number of factors is misspecified. Specifically, we extend the original Stock and Watson framework by introducing four indicators from the price side of the economy, which are expected to share a second latent factor that tracks inflation rather than real activity. By mixing real and price indicators in a single-factor model, we deliberately misspecify the factor structure to illustrate what happens when variables are added blindly under the assumption that a larger dataset inherently yields a better fit.

3.1 Parallel analysis to determine the number of factors

Before estimating the extended model, we determine the number of factors in the original Stock and Watson model.

In nowcasting applications, the optimal number of factors is typically determined using the information criteria proposed by Bai and Ng (2002), which minimize unexplained variance while applying a penalty for each additional factor. However, these criteria are derived for static factor models and is consistent only for $T, N \rightarrow \infty$, where N is the number of variables and T is the time dimension. Indeed, as Bai and Ng note: *“When $\min\{N, T\}$ is 40 or larger, the proposed tests give precise estimates of the number of factors. Since our theory is based on large N and large T , it is not surprising that for very small N or T , the proposed criteria are inadequate.”*

Given that our sample lacks the large cross-sectional dimension required by these criteria and given the underlying assumption of a dynamic factor model, we adopt an alternative approach to determine the appropriate number of factors. We therefore use parallel analysis, originally proposed by Horn (1965) to address a weakness of the Kaiser rule which retains factors with eigenvalues greater than one. Horn argues that in finite samples the eigenvalues of a covariance matrix computed from pure noise are not exactly equal to one because of sampling variation. Therefore, by keeping all factors that exceed the unit cutoff, one usually overestimates the number of factors. Parallel analysis corrects for this by comparing the eigenvalues of the observed data with those obtained from simulated noise data, retaining only factors with an associated eigenvalue larger than the one of the noise benchmark.

In the standard static application, the idiosyncratic component is simulated as white noise. To better adapt this to our dynamic application, we generate noise data from an AR(2) process using a Monte Carlo approach with 500 simulations. For each iteration, we use the same number of observations as our sample (633), applying the coefficients and variances derived from the Kalman filter estimates in Section 1.2. Finally, we average the eigenvalues across all 500 simulations to construct a noise benchmark that accurately takes into account the time-dependence structure of our data.

The results of running parallel analysis on the original Stock and Watson model are displayed in Figure 9. The first eigenvalue has a magnitude of 2.750, well above the AR(2) noise benchmark of 1.090, providing evidence of a dominant common factor driving the co-movements across the four real-activity indicators. All remaining eigenvalues fall below the noise benchmark, with the second one already dropping sharply to around 0.642, well below the noise threshold. Overall, parallel analysis supports the one-factor assumption of the original Stock and Watson model, with the four indicators driven by a single latent factor representing the business cycle.

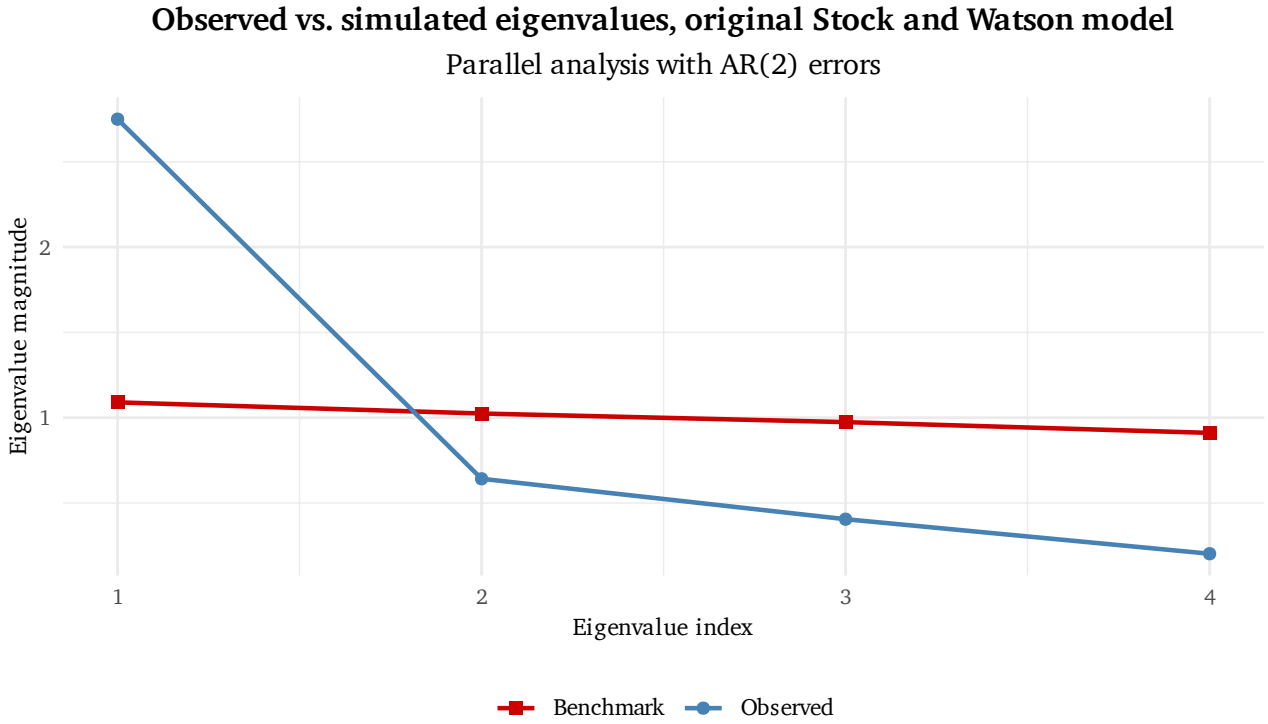


Figure 9: Parallel analysis with AR(2) simulated noise, original Stock and Watson model.

3.2 Introducing price-side indicators

We now extend the dataset by adding four indicators from the price side of the U.S. economy. The four price indicators we add are the following, all expressed in log-differences to reflect percentage changes:

- CPIAUCSL (Consumer Price Index, all items).
- CPILFESL (Consumer Price Index, all items less food and energy).
- PCEPI (Personal Consumption Expenditures Price Index).
- PPIACO (Producer Price Index, all commodities).

Among these, *PCEPI* also serves as the target variable to evaluate the estimated factor's mapping to inflation.

The result of running parallel analysis on the augmented dataset is shown in Figure 10. Notably, both the first and the second observed eigenvalues exceed the AR(2) noise benchmark. The first eigenvalue is approximately 3.158 (versus a benchmark of 1.197), and the second is around 2.606 (against 1.112). All the other eigenvalues drop below the noise benchmark. Hence, this analysis suggests that the four indicators we have added to the dataset share a common latent factor that reflects price pressures in the economy, which is distinct from the real business cycle factor that drives the co-movements of the original four indicators.

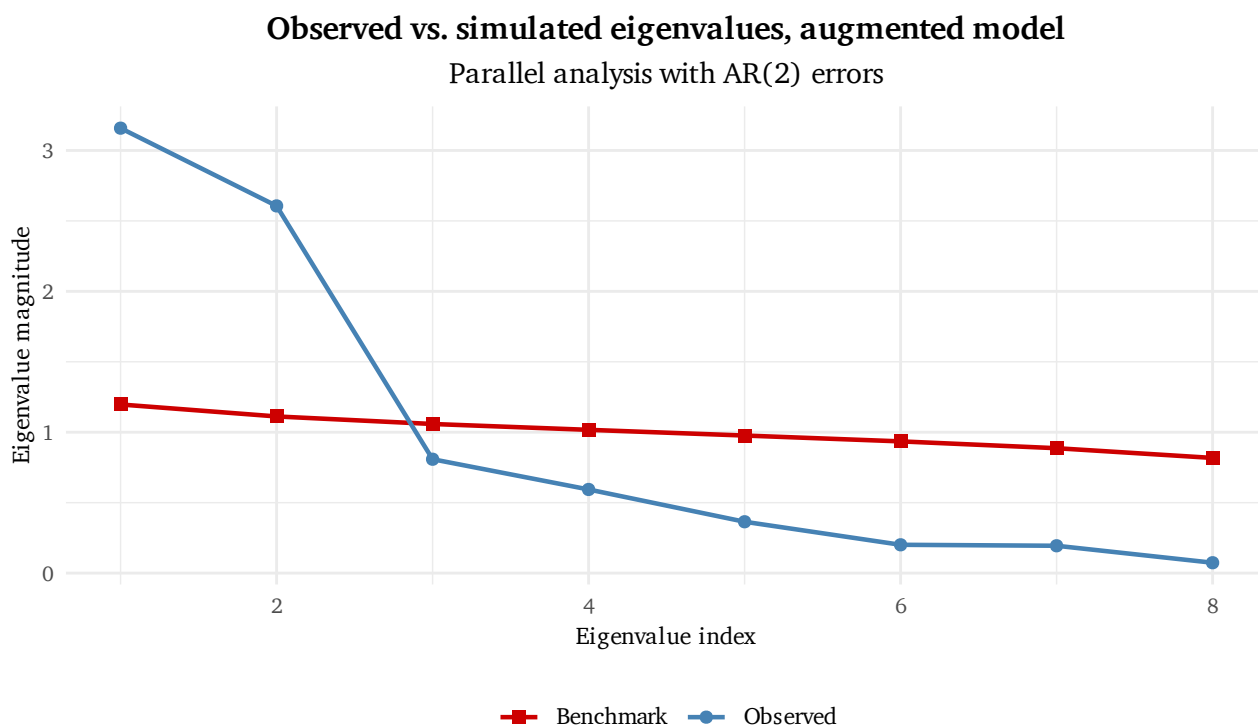


Figure 10: Parallel analysis with AR(2) simulated noise, augmented model.

As shown in Figure 11, the single factor estimated on the augmented eight-indicator dataset continues to track real economic activity, though with worse performance compared to the Stock and Watson model.

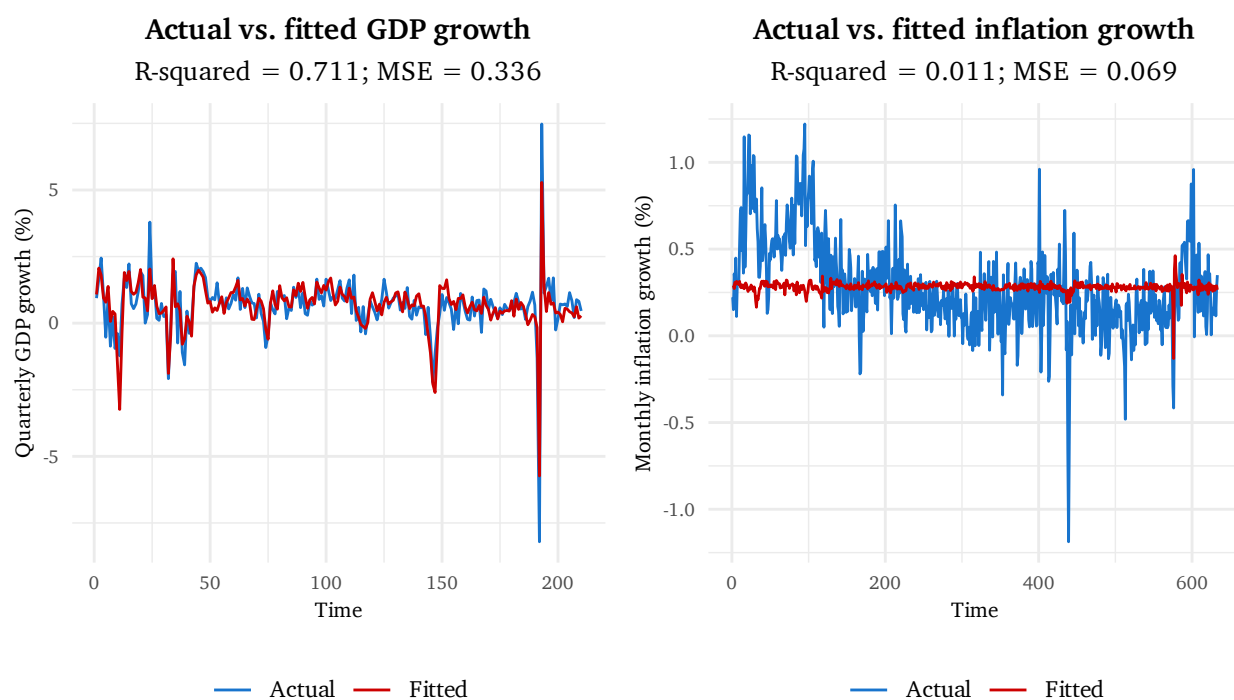


Figure 11: Fitted vs. actual GDP and inflation growth, misspecified augmented model.

Indeed, its fit to actual GDP growth deteriorates to an R^2 of 0.711 and an MSE of 0.336, against the baseline model's R^2 of 0.763 and MSE of 0.276. Interestingly, despite the inclusion of *PCEPI* in the indicator set, the estimated factor fails to capture inflation dynamics altogether, yielding an R^2 of just 0.011 against monthly *PCEPI* growth.

The factor loadings reported in Table 3 make the situation clearer. The four real-activity indicators all receive positive and significant loadings, between 0.484 and 0.967, while the four price indicators are assigned loadings closer to zero. The Kalman filter essentially estimates the original business cycle factor, discarding most of the information brought by the price indicators.

Table 3: Maximum likelihood estimates of the misspecified augmented model.

	Coincident Indicators							
	IPMAN	CMRMTSPL	PAYEMS	W875RX1	CPIAUCSL	CPILFESL	PCEPI	PPIACO
λ_i	0.967	0.688	0.751	0.484	0.101	0.102	0.081	0.157
	(0.027)	(0.030)	(0.034)	(0.035)	(0.029)	(0.022)	(0.026)	(0.036)

Standard errors in parentheses.

To explain this outcome, we draw a parallel to Probabilistic Principal Component Analysis (PPCA). Simplifying to a static cross-sectional case, our measurement equation reduces to $y_t = \lambda f_t + e_t$, or, in matrix notation, $y = \Lambda f + e$. As shown by Tipping and Bishop (1999), when $y \sim N(0, C)$, with $C = \Lambda\Lambda^T + \sigma^2 I$, then the likelihood is maximized when:

$$\Lambda_{ML} = U(D - \sigma^2 I)^{1/2} R$$

where the q column vectors in the $N \times q$ matrix U are the principal eigenvectors of the sample covariance matrix S , with corresponding eigenvalues $d_1, \dots, d_q$ in the $q \times q$ diagonal matrix D , and R is an arbitrary $q \times q$ orthogonal rotation matrix. Under our single-factor specification ($q = 1$), the rotation matrix vanishes and the expression collapses to:

$$\lambda_{ML} = u(d - \sigma^2)^{1/2}$$

Since $u(d - \sigma^2)^{1/2}$ is a scalar, the direction of λ_{ML} is entirely determined by the leading eigenvector u of the sample covariance matrix S , which loads on the block associated to the largest eigenvalue. As reported in Figure 12, we have an approximately two-block sample covariance matrix structure, where the pairwise correlations in the real-activity indicators group range from 0.40 to 0.76, and the ones in the price group range from 0.24 to 0.92. Moreover, the pairwise correlations across groups are all close to zero, ranging from -0.12 to 0.23 , suggesting that the two groups are uncorrelated. Given that the Maximum Likelihood Estimator maximizes whichever direction corresponds to the highest eigenvalue, the Kalman filter in this case recovers the business cycle factor, assigning loadings close to zero to the price indicators and therefore absorbing the price variation into the idiosyncratic error terms.

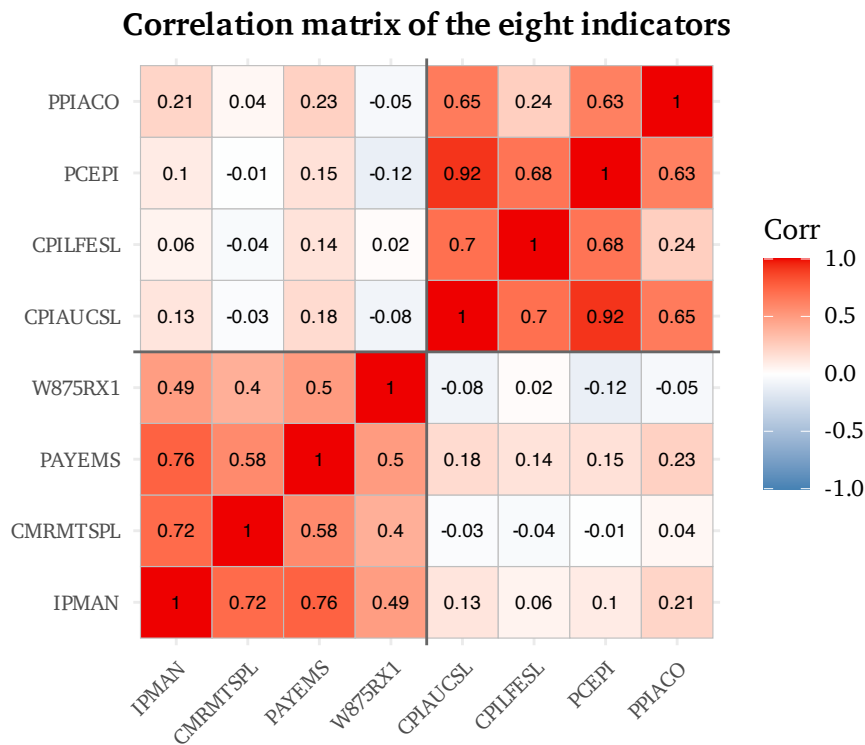


Figure 12: Correlation heatmap of the eight indicators included in the augmented model.

3.3 Re-estimating with the correct specification

As shown in Figure 13, after re-estimating the model under the correct two-factor specification the Kalman filter recovers both latent factors.

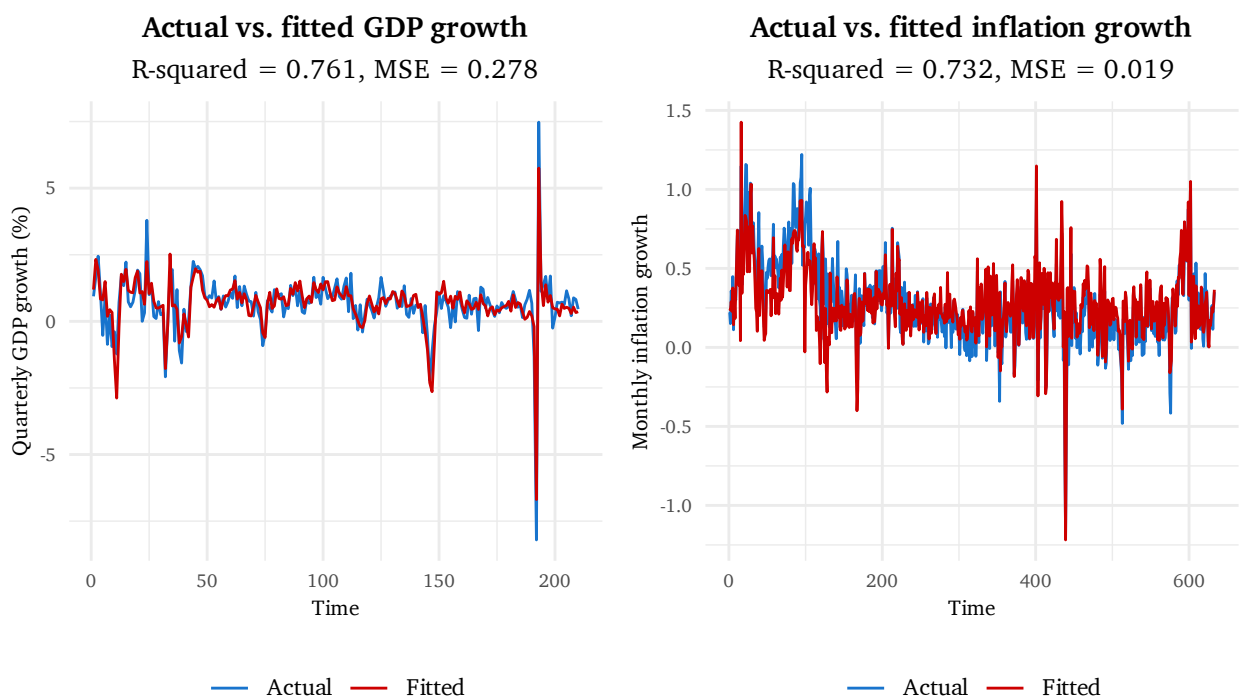


Figure 13: Fitted vs. actual GDP and inflation growth, correctly specified augmented model.

Factor 1 tracks again real economic activity closely, yielding an R^2 of 0.761 and an MSE of 0.278 against quarterly GDP growth, almost identical to the original Stock and Watson model ($R^2 = 0.763$ and $MSE = 0.276$). Meanwhile, *factor 2* now maps well to price pressures, yielding an R^2 of 0.732 and an MSE of 0.019 against monthly *PCEPI* inflation.

As reported in Table 4, *factor 1* indeed loads heavily on the real-activity indicators (between 0.528 and 0.924), while assigning loadings near zero to all four price indicators (between 0.074 and 0.161). Conversely, *factor 2* loads strongly on price indicators (between 0.195 and 0.734) while assigning near-zero loadings to the real-activity ones (between -0.151 and 0.050).

Table 4: Maximum likelihood estimates of the correctly specified two-factor model.

	Coincident Indicators							
	IPMAN	CMRMTSPL	PAYEMS	W875RX1	CPIAUCSL	CPILFESL	PCEPI	PPIACO
$\lambda_{i,1}$	0.924 (0.031)	0.712 (0.031)	0.776 (0.035)	0.528 (0.036)	0.098 (0.030)	0.099 (0.022)	0.074 (0.027)	0.161 (0.037)
$\lambda_{i,2}$	0.000 (0.000)	-0.118 (0.020)	0.050 (0.023)	-0.151 (0.029)	0.734 (0.023)	0.195 (0.023)	0.562 (0.023)	0.562 (0.033)

Standard errors in parentheses. IPMAN Factor 1 fixed to 0.

Note that, to achieve identification, we imposed a lower-triangular restriction on the loading matrix Λ . This is because a factor model $y = \Lambda f + e$ is invariant to orthogonal rotations Q , as:

$$(\Lambda Q)(Q^\top f) = \Lambda(QQ^\top)f = \Lambda I f = \Lambda f$$

Hence, infinitely many rotations of the same factor model produce the same likelihood and the model is unidentified without additional constraints. We therefore set the upper-right triangle of Λ to zero, which for two factors requires setting $\frac{q(q-1)}{2} = \frac{2(2-1)}{2} = 1$ restriction (John Geweke and Zhou 1996). Consistently with this, we set the loading of *IPMAN* on *factor 2* to zero by construction.

4 Conclusion

This project examined the robustness of the Kalman filter as an estimator for dynamic factor models, using the Stock and Watson (1988) framework as our model specification to play with. We subjected the filter to two forms of misspecification: violations of the Gaussianity assumption on the shock distribution, and misspecification of the number of latent factors. Taken together, the results illustrated where the Kalman filter precisely breaks down, how severely, and where it holds strong.

The simulations in Section 2 confirm the theoretical QMLE consistency result: the Kalman filter recovers the latent factor regardless of whether shocks are Gaussian, student-t, or skew-t. All three distributions converge to the same asymptotic factor recovery correlation of approximately 0.7787. However, the cost of misspecification is paid in finite samples. This comes from the fact that the Kalman gain, which updates our estimate for the latent factor based on updated information, is misweighted under fatter tails. In other words, the filter is over-responsive to extreme draws that are more common under a student-t or skew-t distribution; these standard deviations are reported in Table 2, nearly 1.5 times and 2 times the Gaussian baseline for student-t and skew-t, respectively. We also demonstrate in Section 2.3.1 that regardless of the skew value, the factor is recovered in the limit, but there is an increasing cost in volatility in finite sample.

Section 3 illustrates a more severe failure scenario. When the data-generating process contains two latent factors but the model is estimated with only one, the filter does not average across both factors; rather, it selects the direction of maximum variance and discards the rest. In our application, the single-factor model collapses onto the real-activity factor and ignores the price information entirely. Note, however, that this is not a failure of the Kalman filter per se, but a failure of model specification. The filter optimally solves the problem it is given. The parallel analysis in Section 3.1 provides a practical diagnostic for detecting this misspecification before estimation.

The practical implications are that the Kalman filter is surprisingly robust to distributional violations, thanks to the QMLE consistency result, but fragile to factor count misspecification, which is important when trying to fit a model. In practice, diagnosing the number of factors before committing to a model is more important than worrying about whether shocks are exactly Gaussian.

Several extensions are left for future work. First, our simulations assume a known and fixed model structure, including AR(2) factor and errors, and static loadings on the factor; allowing for time-varying loadings would more closely mirror the empirical applications in which the Kalman filter is deployed in central banks (Del Negro and Otrok 2008). Second, we assess factor recovery via correlation with a known true factor, which is only feasible in simulation; in applied settings, the researcher never observes the truth, and alternative evaluation criteria would be needed.

Bibliography

- Bai, Jushan, and Serena Ng. 2002. "Determining the Number of Factors in Approximate Factor Models." *Econometrica* 70 (1): 191–221.
- Del Negro, Marco, and Christopher Otrok. 2008. "Dynamic Factor Models with Time-Varying Parameters: Measuring Changes in International Business Cycles." *Federal Reserve Bank of New York Staff Reports*, no. 326.
- Durbin, James, and Siem Jan Koopman. 2012. *Time Series Analysis by State Space Methods*. 2nd ed. Oxford: Oxford University Press.
- Geweke, J. 1993. "Bayesian Treatment of the Independent Student-t Linear Model." *Journal of Applied Econometrics* 8 (S1): S19–40.
- Geweke, John, and Guofu Zhou. 1996. "Measuring the Pricing Error of the Arbitrage Pricing Theory." *The Review of Financial Studies* 9 (2): 557–87. <https://doi.org/10.1093/rfs/9.2.557>.
- Hamilton, James D. 1994. *Time Series Analysis*. Princeton, New Jersey: Princeton University Press.
- Horn, John L. 1965. "A Rationale and Test for the Number of Factors in Factor Analysis." *Psychometrika* 30 (2): 179–85.
- Kalman, Rudolph Emil. 1960. "A New Approach to Linear Filtering and Prediction Problems." *Transactions of the ASME—Journal of Basic Engineering* 82 (Series D): 35–45.
- Seabold, Skipper, and Josef Perktold. 2010. "Statsmodels: Econometric and Statistical Modeling with Python." <https://doi.org/10.25080/Majora-92bf1922-011>.
- Stock, James H, and Mark W Watson. 1988. "A Probability Model of the Coincident Economic Indicators." Working Paper 2772. National Bureau of Economic Research. <https://www.nber.org/papers/w2772>.
- Tipping, Michael E., and Christopher M. Bishop. 1999. "Probabilistic Principal Component Analysis." *Journal of the Royal Statistical Society: Series B (Statistical Methodology)* 61 (3): 611–22. <https://doi.org/10.1111/1467-9868.00196>.
- White, Halbert. 1982. "Maximum Likelihood Estimation of Misspecified Models." *Econometrica: Journal of the Econometric Society* 50 (1): 1–25.